

Question 1

- (a) Differentiate with respect to x :
- $\sin^{-1} x$
 - $\ln(\tan x)$
- (b) (i) Without the use of calculus, sketch
 $y = (x-1)(x^2 - 4)$
- (ii) Hence, solve the inequality
 $(x-1)(x^2 - 4) < 0$
- (c) Find the acute angle, *to the nearest degree*, between the lines
 $3x - 4y + 8 = 0$ and $x + 2y + 1 = 0$

Question 2

- (a) For what values of x is $\frac{x+4}{x-1} < 6$
- (b) Two chords AB and CD of a circle meet when produced at a point P outside the circle. Prove that triangle ADP and triangle CBP are similar.
- (c) Find the indefinite integrals:
- $\int \frac{x+1}{x^2+4} dx$
 - $\int (1 - \cos^2 x) dx$

Question 3

- (a) Evaluate: $\int_0^1 \frac{x}{\sqrt{1+x}} dx$, *using the substitution* $x = u^2 - 1$
- (b) A spherical balloon leaks air such that the radius decreases at a rate of 5 mm/sec. Calculate the rate of change of the volume of the balloon when the radius is 100 mm.
- (c) AB is the diameter and AC a chord of a circle. The bisector of angle BAC cuts the circle at D. Prove that the tangent at D is perpendicular to AC.

(11)

Question 4

- (a) If $P(x) = x^3 - bx^2 - bx + 4$ is divisible by $(x - 2)$, find the value of "b" and hence all the zeros of $P(x)$.
- (b) If α and β are roots of $x^2 + bx + q = 0$ form the equation, in general form, whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.
- (c) The rate of blood flow [units/sec] through an artery was found *experimentally* to be:

$$r(t) = 0.4 - \sin(\pi t) \quad \text{for} \quad 0 \leq t \leq 2.$$

- (i) What is the total blood that flows over the interval $[0, 2]$
- (ii) It is known that $r(t) = 0$ for $t \approx \frac{1}{6}$, use one step of Newton's Approximation to find an improved root to *two decimal places*.

Question 5

- (a) Differentiate with respect to x : $\tan^{-1}(\cos x)$
- (b) Sketch the function $y = 2 \cos^{-1} \frac{x}{3}$, stating its domain & range.
- (c) One hundred grams of sugar cane in water are being converted into dextrose at a rate which is proportional to the amount at any time i.e., if M grams are converted in t minutes, then $\frac{dM}{dt} = k(100 - M)$ where k is a constant.
Show that $M = 100 + Ae^{-kt}$, where A is a constant, satisfies the differential equation. Find A , given that where $t = 0$, $M = 0$.
If 40 grams are converted in the first 10 minutes, find how many grams are converted in the first 30 minutes.

Question 6

- (a) $P(2ap, ap^2)$ is any point on the parabola $x^2 = 4ay$.
The line k goes through the focus S and is parallel to the tangent at P .
- (i) Find the equation of the line k .
- (ii) The line k intersects the X -axis at Q . Find the equation of the locus of the midpoints of the interval QS and give a precise description of this locus.
- (b) (i) Without differentiation what is the gradient of the line $y = x$ and give your reason?
- (ii) State the **Product Rule for Differentiation**
- (iii) Hence prove by Mathematical Induction:
- $$\frac{d}{dx}(x^n) = nx^{n-1} \quad \text{for all } n \geq 1, n \text{ is an integer.}$$

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Question 7

(a)

A particle moves in such a way that its displacement x cm from the origin O after time t secs is given by:-

$$x = \sqrt{3} \cos 3t - \sin 3t.$$

- (i) Show that the particle moves with Simple Harmonic Motion.
- (ii) Evaluate the period of the motion.
- (iii) Find the time at which the particle first passes through the origin.
- (iv) Find the velocity when the particle is 1 cm from the origin on its first oscillation.

(b) (i) Prove $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

(ii) Prove $\frac{d}{dx} (x \ln x) = 1 + \ln x$

- (iii) The acceleration of a particle moving in a straight line and starting at 1 cm on the positive side of the origin, at rest is given by:

$$\frac{d^2x}{dt^2} = 1 + \ln x$$

derive the equation relating v and x .
Hence evaluate v when $x = e^2$

End of Examination

Q1(a) (i) $y = \sin^{-1} x$

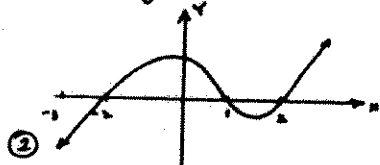
① $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

(ii) $y = \ln(\tan x)$

$\frac{dy}{dx} = \frac{\sec^2 x}{\tan x}$

$= \frac{1}{\sin x \cos x}$

(b) (i) $y = (x-1)(x^2-4)$



$(x-1)(x^2-4) < 0$

② $\{x: x < -2 \text{ or } 1 < x < 2\}$

(c) ①: $3x - 4y + 8 = 0$ $m_1 = \frac{3}{4}$

②: $x + 2y + 1 = 0$ $m_2 = -\frac{1}{2}$

Let ϕ = acute angle between l_1 & l_2

$\therefore \tan \phi = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{\frac{3}{4} + \frac{1}{2}}{1 - \frac{3}{8}} \right|$

$= \left| \frac{\frac{5}{4}}{\frac{5}{8}} \right|$

$\tan \phi = 2$

$\therefore \phi = 63^\circ 26'$

③ $\phi = 63^\circ$ (To nearest degree)

Q2(a) $\frac{x+4}{x-1} < 6$

$x-1 \neq 0 \therefore x \neq 1$

Then $\frac{x+4}{x-1} < 6$

$x+4 < 6x-6$

$10 < 5x$

$x > 2$



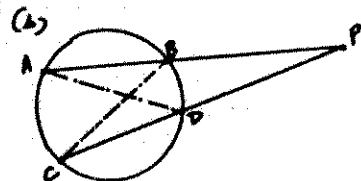
Test points from regions:—

$x=0: -4 < 6$ ✓

$x=1: 11 < 6$ ✗

$x=3: 7 < 6$ ✓

④: $\{x: x < 1 \text{ or } x > 2\}$

Aim: To prove $\triangle ADP \parallel \triangle BCP$.

Data: AB & BC are chords meeting externally at P.

Proof: In $\triangle ADP$, $\triangle BCP$ —

① $\angle APB = \angle BPC$ (Common $\angle$).

② $\angle DAP = \angle BCP$ (Angles standing on same arc).

③ $\angle ADP = \angle BCP$ (3rd angle) [∴ $\triangle ADP \sim \triangle BCP$].

④ $\therefore \triangle ADP \parallel \triangle BCP$ (3 angles).

(c) (i) $\int \frac{x+1}{x^2+2x} dx = \frac{1}{2} \int \frac{2x}{x^2+2x} dx + \frac{1}{2} \int \frac{1}{x^2+2x} dx$

② $= \frac{1}{2} \ln(x^2+2x) + \frac{1}{2} \ln \frac{x}{x+2} + c$

(ii) $\int (1 - \cos^2 x) dx$

$= \int \sin^2 x dx$

$= \frac{1}{2} \int (1 - \cos 2x) dx$

③ $= \frac{1}{2} [x - \frac{1}{2} \sin 2x] + c$

Q3(a) $\int \frac{x}{\sqrt{1+x}} dx$

⑤

$= \int \frac{u^2-1}{\sqrt{u^2-1}} 2u du$

$= \int \frac{u^2-1}{u} 2u du$

$= 2 \left(\frac{u^3}{3} - u \right) \Big|_1^{\sqrt{2}}$

$= 2 \left(\frac{2\sqrt{2}}{3} - \sqrt{2} - \frac{1}{3} + 1 \right)$

$= 2 \left(\frac{2-\sqrt{2}}{3} \right)$

$x = u^2 - 1$

$\frac{dx}{2u} = 2u du$

$dx = 2u du$

At $x=0, u^2-1=1$

$u = \pm 1$

At $x=1, 1 = u^2 - 1$

$2 = u^2$

$u = \pm \sqrt{2}$

(b) $\frac{dr}{dt} = 5 \text{ mm/s}$

⑤ $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

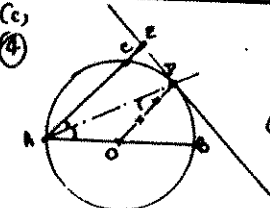
But $V = \frac{4}{3} \pi r^3$

$\frac{dV}{dr} = 4\pi r^2$

At $r = 100 \text{ mm}$

$\frac{dV}{dt} = 4\pi (100)^2 \cdot 5$

$= 200000\pi \text{ mm}^3/\text{s}$

Aim: To prove $AE \perp ED$.Given: AD bisects $\angle BAC$.

Construction: OD and extend AC to tangent at E.

Proof: $\triangle OAD$ is isosceles = radii & $OD \perp AC$ (Data). $\therefore \angle ODA = \angle OAD = \angle DAC$ (Data).But $\angle CAD$ & $\angle ODA$ are alternate angles in $\parallel$ lines AC & OE . $\therefore OD \perp OE$ at D. $\therefore AC \perp OE$ at E.

Q4(a) $P(x) = x^3 - 6x^2 - 6x + 4$ ②

④ $P(2) = 8 - 46 - 12 + 4 = 0$

$-66 = -12$

$b = 2$

$\therefore P(x) = x^3 - 2x^2 - 2x + 4$

$$\begin{array}{r} x^3 - 2x^2 - 2x + 4 \\ x-2 \overline{) x^3 - 2x^2 - 2x + 4} \\ \underline{x^3 - 2x^2} \\ -2x + 4 \\ \underline{-2x + 4} \\ 0 \end{array}$$

$\therefore P(x) = (x-2)(x^2-2)$

$= (x-2)(x+\sqrt{2})(x-\sqrt{2})$

$\therefore P(x) = 0$

$(x-2)(x+\sqrt{2})(x-\sqrt{2}) = 0$

$x = \pm \sqrt{2}, 2$

③ (b) $x^2 + 6x + 9 = 0$

$\alpha + \beta = -6$

$\alpha\beta = 9$

$\therefore x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right) x + \frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 0$

$\Rightarrow x^2 - \left(\frac{\alpha^2 + \beta^2}{\alpha\beta} \right) x + 1 = 0$

$x^2 - \left(\frac{(-6)^2 - 2(9)}{9} \right) x + 1 = 0$

$x^2 - \left(\frac{36 - 18}{9} \right) x + 1 = 0$

$x^2 - \left(\frac{18}{9} \right) x + 1 = 0$

$x^2 - 2x + 1 = 0$

(c) $r(t) = 0.4 - \sin(\pi t)$

Total blood:

(i) $\int_0^2 r(t) dt$

$= \int_0^2 (0.4 - \sin \pi t) dt$

$= 0.4t + \frac{1}{\pi} \cos \pi t \Big|_0^2$

$= 0.8 + \frac{\cos 2\pi}{\pi} - \frac{\cos 0}{\pi}$

$= 0.8 + \frac{1}{\pi} - \frac{1}{\pi}$

$= 0.8 \text{ units of blood}$

(ii) hypotenuse not z_1 :

$$z_2 = z_1 - \frac{r(y_1)}{r(y)}$$

$$r(\frac{1}{2}) = 0.4 - \sin \frac{\pi}{6}$$

$$= 0.4 - \frac{1}{2}$$

$$= -0.1$$

$$r'(\frac{1}{2}) = \pi \cos \frac{\pi}{6}$$

$$= \frac{\pi \sqrt{3}}{2}$$

$$z_2 = \frac{1}{2} - \left[\frac{-0.1}{\frac{\pi \sqrt{3}}{2}} \right]$$

$$= 0.13 \quad (\angle 200^\circ)$$

Q6(a) $y = \tan^{-1}(\cos x)$.

Let $u = \cos x$, $y = \tan^{-1}u$

$$\frac{du}{dx} = -\sin x, \quad \frac{dy}{du} = \frac{1}{1+u^2}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{1+u^2} \times -\sin x$$

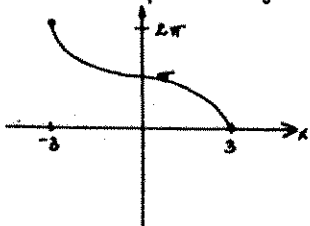
$$= \frac{-\sin x}{1+\cos^2 x}$$

(b) $y = 2\cos^{-1}\frac{x}{3}$

$$-1 \leq \frac{x}{3} \leq 1 \quad 0 \leq y \leq \pi$$

$$-1 \leq \frac{x}{3} \leq 1 \quad 0 \leq \frac{y}{2} \leq \pi$$

$$-3 \leq x \leq 3 \quad 0 \leq y \leq 2\pi$$



(c) $M = 100 + Ae^{-kt}$

$$\frac{dM}{dt} = -Ake^{-kt}$$

$$= k(100 - (100 + Ae^{-kt}))$$

$$\frac{dM}{dt} = k(100 - M)$$

When $t=0$, $M=0$:

$$0 = 100 + Ae^0$$

$$A = -100$$

$$\therefore M = 100 - 100e^{-kt}$$

$$\& \text{ At } t = 100, M = 100e^{-10k}$$

$$-60 = -100e^{-10k}$$

$$e^{-10k} = 0.6$$

$$-10k \ln e = \ln 0.6$$

$$k = \frac{\ln 0.6}{-10}$$

$$k = 0.05103$$

When $t=30$:

$$M = 100 - 100e^{-1.5309}$$

$$\therefore M = 78.4 \text{ grams converted!}$$

Q6(b) Gradient of k :

(a) $u_k = p$

(i) eq: of k :

$$y - a = p(x - b)$$

$$(ii) y = px + a$$

Gradients of Q :

$$(i) y=0 \Rightarrow x = -\frac{a}{p}$$

$$\therefore Q = (-\frac{a}{p}, 0)$$

Midpoint of QS :

$$X = \frac{-\frac{a}{p} + 0}{2} \quad Y = \frac{0 + a}{2}$$

$$X = -\frac{a}{2p} \quad Y = \frac{a}{2}$$

$$X = -\frac{a}{2p} \quad Y = \frac{a}{2}$$

$$\therefore Q = (-\frac{a}{2p}, \frac{a}{2})$$

Locus is straight line $y = \frac{a}{2}$

(as it is independent of p).

(b) (i) $y = x$

(1) has a gradient of 1

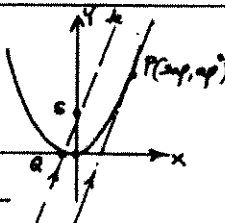
because:

(2) line makes an angle of 45°

with X -axis $\tan 45^\circ = 1$.

or (3) in form $y = mx + c \Rightarrow m = 1$

or (4) 15° from Y -axis.



(i) Product Rule for $y = uv$
where u & v are functions of x .

$$\therefore \frac{d}{dx}(uv) = v \frac{du}{dx} + u \frac{dv}{dx}$$

(iii) Step 1 To prove true for $n=1$.

$$\frac{d}{dx}(x^1) = 1x^0 = 1$$

& gradient of line $y=x$ is 1.

$\therefore$ True for $n=1$.

(ii) Assume true for $n=k$.

$$\frac{d}{dx}(x^k) = kx^{k-1}$$

(iii) prove true for $n=k+1$

$$\therefore \frac{d}{dx}(x^{k+1}) = (k+1)x^k$$

$$\text{As } \frac{d}{dx}(x^{k+1}) = \frac{d}{dx}(x^k \cdot x)$$

$$= \left[\frac{d}{dx}(x^k) \right] \cdot x + \left[\frac{d}{dx}(x) \right] x^k$$

$$= kx^{k-1} \cdot x + x^k$$

$$= kx^k + x^k$$

$$= (k+1)x^k \quad \text{QED.}$$

(iv) If true for $n=1$, then true for $n=1+1=2$ and so on for all positive integral values of n .

Q7(a) (i), $x = \sqrt{3} \cos 3t - \sin 3t$

$$\frac{dx}{dt} = -3\sqrt{3} \sin 3t - 3 \cos 3t$$

$$\frac{d^2x}{dt^2} = -9\sqrt{3} \cos 3t + 9 \sin 3t$$

$$= -9(\sqrt{3} \cos 3t - \sin 3t)$$

$$\ddot{x} = -9x \quad (\lambda^2 = 9)$$

(ii) Period of motion $T = \frac{2\pi}{\omega}$

$$\therefore T = \frac{2\pi}{3}$$

$$(iii) \sqrt{3} \cos 3t - \sin 3t = 0$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$2\left(\frac{\sqrt{3}}{2} \cos 3t - \frac{1}{2} \sin 3t\right) = 0$$

$$2 \cos(3t + \frac{\pi}{6}) = 0$$

$$\cos(3t + \frac{\pi}{6}) = 0$$

$$3t + \frac{\pi}{6} = \frac{\pi}{2}$$

$$t = \frac{\pi}{9}$$

(iv) (i) when $2 \cos(3t + \frac{\pi}{6}) = 1$

$$\cos(3t + \frac{\pi}{6}) = \frac{1}{2}$$

$$3t + \frac{\pi}{6} = \frac{\pi}{3}$$

$$3t = \frac{\pi}{6}$$

$$t = \frac{\pi}{18}$$

$$\text{At } t = \frac{\pi}{18}, v = \frac{dx}{dt}$$

$$= -3\sqrt{3} \sin \frac{3\pi}{18} - 3 \cos \frac{3\pi}{18}$$

$$= -3\sqrt{3} \sin \frac{\pi}{6} - 3 \cos \frac{\pi}{6}$$

$$= -\frac{3\sqrt{3}}{2} - \frac{3\sqrt{3}}{2}$$

$$= -\frac{6\sqrt{3}}{2}$$

$$v = -3\sqrt{3} \text{ cm/sec.}$$

(b) (i) $\frac{dv}{dt} = \frac{d}{dt}(\frac{1}{2}v^2)$

$$\frac{d}{dx}(\frac{1}{2}v^2) = \frac{d}{dv}(\frac{1}{2}v^2) \frac{dv}{dx}$$

$$= v \frac{dv}{dx}$$

$$= \frac{dx}{dt} \cdot \frac{dv}{dx}$$

$$= \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

$$(ii) \frac{d}{dx}(x \ln x) = 1 + \ln x$$

$$\frac{d}{dx}(x) \ln x + \left[\frac{d}{dx}(\ln x) \right] [x]$$

$$= 1 \ln x + \frac{1}{x} \cdot x$$

$$= \ln x + 1 \quad \text{①}$$

$$(iii) \int \frac{d}{dx}(x \ln x) dx = \int (1 + \ln x) dx$$

$$\frac{1}{2}v^2 = \int \frac{d}{dx}(x \ln x) dx$$

$$\frac{1}{2}v^2 = x \ln x + x$$

$$v=0, x=1 \Rightarrow \frac{1}{2}v^2 = 0$$

$$v = \frac{d^2x}{dt^2} \ln x$$

$$\text{At } x=e, v = \frac{d^2x}{dt^2} \ln e = 2e \text{ cm/s}$$

$$\text{or } \cos 3t = \frac{1}{2}$$

$$\tan 3t = \sqrt{3}$$

$$3t = \frac{\pi}{3}$$

$$t = \frac{\pi}{9}$$